



CAIRO UNIVERSITY

FACULTY OF SCIENCE

PHYSICS DEPARTMENT

Equations Useful for Undergrad Physics Students

Author: Abdelrahman M. ABDELMAGID

Contents

1	ABSTRACT	2
2	VECTORS	2
2.1	LINE ELEMENTS	2
3	TRIGONOMETRY	2
3.1	Trigonometric Differentiation	4
4	HYPERBOLIC FUNCTIONS	4
5	DIFFERENTIATION	5
6	Expantions	5
7	Functional	6
8	Maxwell Equation	6
9	Gamma Functions	7
10	Essential Series	8

1 ABSTRACT

These are the Equations and Relations that i needed during studying in the Physics Departement as an undergrad student, I had problems memorizing most of these.

2 VECTORS

$$\vec{A} \times \vec{B} \times \vec{C} = \vec{B}(A.C) - \vec{C}(A.B) \quad (1)$$

2.1 LINE ELEMENTS

in Cartisian co-ordinates:

$$dl^2 = dx^2 + dy^2 + dz^2 \quad (2)$$

in Cylindrical co-ordinates:

$$dl^2 = dr^2 + r^2 d\phi^2 + dz^2 \quad (3)$$

in spherical Co-ordiantes

$$dl^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (4)$$

3 TRIGONOMETRY

$$\sin(\alpha \pm \beta) = \sin(\alpha)\cos(\beta) \pm \cos(\alpha)\sin(\beta) \quad (5)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha)\cos(\beta) \mp \sin(\alpha)\sin(\beta) \quad (6)$$

$$\sin(\alpha)\cos(\beta) = \frac{1}{2}[\sin(\alpha + \beta) + \sin(\alpha - \beta)] \quad (7)$$

$$\cos(\alpha)\cos(\beta) = \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)] \quad (8)$$

$$\sin(\alpha)\sin(\beta) = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)] \quad (9)$$

$$\sin^2 \alpha = \frac{1}{2}[1 - \cos 2\alpha] \quad (10)$$

$$\cos^2 \alpha = \frac{1}{2}[1 + \cos 2\alpha] \quad (11)$$

$$2\sin x \cos x = \sin 2x \quad (12)$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i} \quad (13)$$

$$\cos x = \frac{e^{ix} + e^{-ix}}{2} \quad (14)$$

$$\int_0^\pi \sin\left(\frac{n\pi}{a}x\right) \sin\left(\frac{n'\pi}{a}x\right) dx = \delta_{nn'} \quad (15)$$

3.1 Trigonometric Differentiation

$$\frac{d}{dx} \sin x = \cos x \quad (16)$$

$$\frac{d}{dx} \cos x = -\sin x \quad (17)$$

$$\frac{d}{dx} \tan x = \sec^2 x \quad (18)$$

$$\frac{d}{dx} \csc x = -\csc x \cot x \quad (19)$$

$$\frac{d}{dx} \sec x = \sec x \tan x \quad (20)$$

$$\frac{d}{dx} \cot x = -\csc^2 x \quad (21)$$

4 HYPERBOLIC FUNCTIONS

$$\sinh x = \frac{e^x - e^{-x}}{2} \quad (22)$$

$$\cosh x = \frac{e^x + e^{-x}}{2} \quad (23)$$

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (24)$$

$$\frac{\partial}{\partial x}(Cosh(x)) = Sinh(x) \quad (25)$$

$$\frac{\partial}{\partial x}(Sinh(x)) = Cosh(x) \quad (26)$$

$$\frac{\partial}{\partial x}(Tanh(x)) = Sech^2(x) \quad (27)$$

$$\frac{\partial}{\partial x}(Sech(x)) = -Sech(x)Tanh(x) \quad (28)$$

$$\frac{\partial}{\partial x}(Csch(x)) = -Coth(x)Csch(x) \quad (29)$$

$$\frac{\partial}{\partial x}(Coth(x)) = -Csch^2(x) \quad (30)$$

5 DIFFERENTIATION

$$y = \frac{u}{v} \longrightarrow \frac{dy}{dx} = \frac{vu' - uv'}{v^2} \quad (31)$$

Trigonometric Differentiation

6 Expantions

Theorem 1: Theorem Name

A theorem.

7 Functional

$$\frac{\delta F}{\delta f(x)} = \lim_{\varepsilon \rightarrow 0} \frac{F[f(x) + \varepsilon \delta(x - x')] - F[f(x)]}{\varepsilon} \quad (32)$$

$$J[x] = \int f^p(x) \phi(x) dx \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = P[f(x_o)]^{p-1} \phi(x_o) \quad (33)$$

$$H[f(x)] = \int g[f(x)] dx \quad \longrightarrow \quad \frac{\delta H}{\delta f(x_o)} = g'(f(x_o)) \quad (34)$$

$$J[f] = \int g(f'(y)) dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = -\frac{1}{dx} \left(\frac{d g(f')}{d f'(x)} \right) \quad (35)$$

$$J[f] = \int \left(\frac{\partial f}{\partial y} \right)^2 dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = -2 \frac{\partial^2 f}{\partial y^2} \quad (36)$$

$$G[f] = \int g(y, f) dy \quad \longrightarrow \quad \frac{\delta H}{\delta f(x_o)} = \frac{\partial g(x, f)}{\partial f(x)} \quad (37)$$

$$J[f] = \int g(y, f, f', f'') dy \quad \longrightarrow \quad \frac{\delta J}{\delta f(x_o)} = \frac{\partial g}{\partial f} - \frac{d}{dx} \left(\frac{\partial g}{\partial f'(x)} \right) + \frac{d^2}{dx^2} \left(\frac{\partial g}{\partial f''} \right) \quad (38)$$

8 Maxwell Equation

$$(39)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (40)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (41)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \frac{1}{c^2} \frac{\partial \mathbf{E}}{\partial t} \quad (42)$$

9 Gamma Functions

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt \quad (43)$$

$$\Gamma(n) = (n-1)! \quad (44)$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad (45)$$

$$\int \ln(x) dx = x \ln(x) - x \ln N! = N \ln N - N \quad (46)$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}} \quad (47)$$

$$\int_{-\infty}^{\infty} x^n e^{-\alpha x^2} = \begin{cases} \frac{1}{2} \left(\frac{1}{\alpha}\right)^{n+1} n! & n = 1, 2, 3, \dots \\ \frac{1}{2} \left(\frac{1}{\alpha}\right)^{\frac{2n+1}{2}} \Gamma\left(n + \frac{1}{2}\right) & n = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots \end{cases} \quad (48)$$

10 Essential Series

$$\sum_{n=0}^{N-1} e^{inx} = \frac{1 - e^{iNx}}{1 - e^{ix}} \quad (49)$$